ModelIt!

Student Activity Pack

Climate-Ecosystem Coupled Models

Modeling Feedback Loops Between Climate Change, Ecosystems, and Human Society

Grade 09L3 | NGSS HS-ESS3-5, HS-LS2-6 | Stability and Change / Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think happens when Arctic permafrost melts — and why might this create a 'snowball effect' that makes climate change worse?**

2. **Why do you think some regions of the world will be affected by climate change much more severely than others?**

3. **Draw a diagram showing how you think CO2 emissions lead to rising sea levels — include as many intermediate steps as you can.**

4. **What is a 'tipping point' in the climate system and why should we be concerned about crossing one?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Atmospheric CO2
- Global Temperature
- Ocean Absorption
- Permafrost Methane Release
- Forest Carbon Sequestration
- Species Migration Rate
- Crop Yield Change
- Sea Level Rise
- Extreme Weather Frequency
- Societal Adaptation Capacity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Paris Compliance

Limited warming to 1.5-2.0 degrees C

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Business-As-Usual

High emission trajectory, 3-4 degrees C warming

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Tipping Point Search

Gradual temperature increase

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Albedo Feedback — The reflectivity change caused by melting ice and snow cover — ice reflects 80-90% of solar radiation while open ocean absorbs 90%, creating a powerful positive feedback where ice loss causes more warming which causes more ice loss, particularly important in the Arctic where summer sea ice has declined 40% since 1979

Ocean Acidification — The decrease in ocean pH caused by dissolved CO2 reacting with water to form carbonic acid — currently 30% more acidic than pre-industrial levels, threatening coral reefs, shellfish, and marine food webs, with cascading effects on ocean carbon absorption capacity and marine ecosystem stability

Climate Migration Pressure — The displacement of human populations by climate impacts including sea level rise, extreme heat, drought, and agricultural collapse — currently 20+ million climate-displaced people annually, projected to reach 200+ million by 2050, creating political, economic, and humanitarian cascading effects

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Climate change doesn't just heat the planet — it triggers cascading feedback loops through every ecosystem on Earth. When permafrost melts and releases methane, which accelerates warming, which melts more permafrost, which releases more methane — how do you model a system where the output becomes the input? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the difference between positive feedback loops (self-amplifying) and negative feedback loops (self-correcting) in the climate system — and why are the positive feedbacks winning?**

2. **What did the tipping point analysis reveal about the relationship between Global Temperature and Permafrost Methane Release — and why is this threshold so consequential?**

3. **Why does Forest Carbon Sequestration potentially flip from a negative feedback (absorbing CO_2) to a positive feedback (releasing CO_2) above a certain temperature threshold?**

4. **How would adding Climate Migration Pressure to your model change the analysis of Societal Adaptation Capacity?**

5. **What does the model reveal about climate justice — the relationship between who caused the emissions and who bears the consequences?**

8 STEM Challenge: Design a Regional Climate Adaptation and Mitigation Plan

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YOUR ENGINEERING MISSION

Design an integrated climate adaptation and mitigation strategy for a specific vulnerable region, using model-derived evidence to prioritize interventions and project outcomes.

THE SCENARIO

A coastal city of 2 million people is projected to face 0.5-1.5 meters of sea level rise, a 40% increase in extreme weather events, and significant agricultural disruption by 2080. The city has hired your climate planning team to develop a 50-year integrated strategy that combines mitigation (reducing the city's carbon footprint), adaptation (preparing for unavoidable impacts), and resilience (ensuring recovery from extreme events). Your plan must be evidence-based, economically feasible, and equitable.

Engineering Design Thinking

1. WOULD YOU INVEST MORE IN HARD INFRASTRUCTURE (SEA WALLS, FLOOD BARRIERS) OR NATURE-BASED SOLUTIONS (MANGROVE RESTORATION, GREEN INFRASTRUCTURE) — AND WHAT DOES YOUR MODEL PREDICT ABOUT THE LONG-TERM EFFECTIVENESS OF EACH?

2. HOW WOULD YOU DESIGN THE PLAN TO REMAIN EFFECTIVE ACROSS THE RANGE OF POSSIBLE WARMING SCENARIOS (1.5 DEGREES C TO 4.0 DEGREES C)?

3. WHAT MONITORING AND EARLY WARNING SYSTEMS WOULD YOU INSTALL TO DETECT WHEN TIPPING POINTS ARE BEING APPROACHED?

4. HOW WOULD YOU FUND THE PLAN — CARBON TAXES, GREEN BONDS, FEDERAL DISASTER PREPAREDNESS FUNDING — AND HOW DOES THE FINANCING STRATEGY AFFECT EQUITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists and Earth System Modelers develop and run coupled climate-ecosystem models to project future climate conditions and ecosystem impacts. They work for government agencies (NOAA, NASA, EPA), national laboratories (NCAR, ORNL, PNNL), international organizations (IPCC, WMO), and university research centers, earning $80,000-$170,000/year. Climate Adaptation Planners who design community resilience strategies earn $70,000-$140,000/year.